

**BPS-17 Lecturer Mathematics AJKPSC/FPSC/PPSC/KPSC |
BPS-17 | Mathematics Lecturer Examination**

TOPOLOGY

Comprehensive Study Plan & MCQ Bank

Applied Mathematics — Zamir Hussain

Topics Covered:

Topological Spaces · Open & Closed Sets · Neighbourhoods
Basis & Sub-basis · Interior, Closure, Boundary
Subspace Topology · Product Topology · Quotient Topology
Continuity & Homeomorphism · Topological Equivalence
Separation Axioms: T_0, T_1, T_2, T_3, T_4 (Hausdorff, Regular, Normal)
Compactness · Tychonoff Theorem · Heine-Borel Theorem
Connectedness · Path-Connectedness · Components
Metric Spaces · Completeness · Baire Category Theorem

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1. UNIT 1: TOPOLOGICAL SPACES

Core Definition: Topological Space

A **topological space** is a pair $(X, \mathcal{T})$ where X is a non-empty set and $\mathcal{T}$ is a collection of subsets of X (called **open sets**) satisfying:

1. $\emptyset \in \mathcal{T}$ and $X \in \mathcal{T}$ (empty set and whole set are open)
2. Arbitrary unions: if $\{U_\alpha\} \subseteq \mathcal{T}$, then $\bigcup_\alpha U_\alpha \in \mathcal{T}$
3. Finite intersections: if $U_1, U_2, \dots, U_n \in \mathcal{T}$, then $\bigcap_{i=1}^n U_i \in \mathcal{T}$

The collection $\mathcal{T}$ is called a **topology** on X .

LaTeX Code

```
(X, \mathcal{T}) \quad \emptyset \in \mathcal{T}, \quad X \in \mathcal{T} \\
\bigcup_{\alpha} U_{\alpha} \in \mathcal{T}, \quad \bigcap_{i=1}^n U_i \in \mathcal{T}
```

1.1 Standard Topologies

Important Standard Topologies

- **Indiscrete (Trivial) Topology:** $\mathcal{T} = \{\emptyset, X\}$ — coarsest topology
- **Discrete Topology:** $\mathcal{T} = \mathcal{P}(X)$ (all subsets) — finest topology
- **Standard Topology on $\mathbb{R}$:** Open sets are arbitrary unions of open intervals (a, b)
- **Cofinite Topology:** $U \in \mathcal{T}$ iff $U = \emptyset$ or $X \setminus U$ is finite
- **Cocountable Topology:** $U \in \mathcal{T}$ iff $U = \emptyset$ or $X \setminus U$ is countable
- **Lower Limit Topology (Sorgenfrey Line):** basis $\{[a, b) : a < b, a, b \in \mathbb{R}\}$

Comparison of Topologies

If $\mathcal{T}_1$ and $\mathcal{T}_2$ are topologies on X :

$$\mathcal{T}_1 \subseteq \mathcal{T}_2 \implies \mathcal{T}_1 \text{ is } \mathbf{coarser} \text{ (weaker), } \mathcal{T}_2 \text{ is } \mathbf{finer} \text{ (stronger)}$$

Ordering: Indiscrete $\subseteq$ Any Topology $\subseteq$ Discrete

1.2 Basis for a Topology

Basis and Sub-basis

A collection $\mathcal{B} \subseteq \mathcal{P}(X)$ is a **basis** for a topology if:

1. For each $x \in X$, there exists $B \in \mathcal{B}$ with $x \in B$

2. If $x \in B_1 \cap B_2$ for $B_1, B_2 \in \mathcal{B}$, then $\exists B_3 \in \mathcal{B}$ with $x \in B_3 \subseteq B_1 \cap B_2$

The topology **generated** by $\mathcal{B}$: $U \in \mathcal{T}$ iff U is a union of elements of $\mathcal{B}$.

A collection $\mathcal{S}$ is a **sub-basis** if $\mathcal{B} = \{\text{finite intersections of elements of } \mathcal{S}\}$ forms a basis.

LaTeX Code

```
\mathcal{B}, \quad x \in B_3 \subseteq B_1 \cap B_2, \quad \mathcal{T} = \{U : \\ U = \bigcup B_\alpha\}
```

1.3 Interior, Closure, and Boundary

Interior, Closure, Boundary Formulas

For a subset $A \subseteq X$ in topological space $(X, \mathcal{T})$:

$$\mathring{A} = \bigcup \{U \in \mathcal{T} : U \subseteq A\} \quad (\text{largest open set in } A)$$

$$\overline{A} = \bigcap \{F : F \text{ closed, } A \subseteq F\} \quad (\text{smallest closed set containing } A)$$

$$\partial A = \overline{A} \setminus \mathring{A} = \overline{A} \cap \overline{X \setminus A}$$

Limit point: x is a limit point of A if every open set containing x meets $A \setminus \{x\}$.

Derived set: $A' = \{\text{limit points of } A\}$

$$\overline{A} = A \cup A'$$

LaTeX Code

```
\mathring{A} = \bigcup \{U \in \mathcal{T} : U \subseteq A\} \\ \overline{A} = \bigcap \{F : F \text{ closed, } A \subseteq F\} \\ \partial A = \overline{A} \setminus \mathring{A}
```

Example: Interior, Closure, Boundary in $\mathbb{R}$

Let $A = (0, 1]$ in $(\mathbb{R}, \mathcal{T}_{\text{std}})$:

$$\mathring{A} = (0, 1), \quad \overline{A} = [0, 1], \quad \partial A = \{0, 1\}$$

Let $A = \mathbb{Q}$ (rationals):

$$\mathring{\mathbb{Q}} = \emptyset, \quad \overline{\mathbb{Q}} = \mathbb{R}, \quad \partial \mathbb{Q} = \mathbb{R}$$

1.4 MCQs: Topological Spaces

Q1. Which of the following is a topology on $X = \{a, b, c\}$?

(A) $\{\emptyset, \{a\}, \{b, c\}\}$

(C) $\{\emptyset, \{a, b\}, \{b, c\}, X\}$

(B) $\{\emptyset, \{a\}, \{a, b\}, X\}$

(D) $\{\emptyset, \{a\}, \{b\}, X\}$

✓ **Answer: (B)** Check axioms: $\emptyset, X \in \mathcal{T}$; unions: $\{a\} \cup \{a, b\} = \{a, b\} \in \mathcal{T}$; intersections: $\{a\} \cap \{a, b\} = \{a\} \in \mathcal{T}$. All satisfied. Option (C) fails: $\{a, b\} \cap \{b, c\} = \{b\} \notin \mathcal{T}$.

Q2. In discrete topology on X , every subset is:

- | | |
|-----------------|-----------------------------------|
| (A) Closed only | (C) Neither open nor closed |
| (B) Open only | (D) Both open and closed (clopen) |

✓ **Answer: (D)** In discrete topology $\mathcal{T} = \mathcal{P}(X)$: every subset A is open ($A \in \mathcal{T}$) and its complement is also open, so A is also closed. Every set is **clopen**.

Q3. The indiscrete topology on X has how many open sets?

- | | |
|-------|---------------|
| (A) 1 | (C) $2^{ X }$ |
| (B) 2 | (D) $ X $ |

✓ **Answer: (B)** The indiscrete (trivial) topology is $\mathcal{T} = \{\emptyset, X\}$, which contains exactly **2** open sets regardless of $|X|$.

Q4. Interior of $A = \{1/n : n \in \mathbb{N}\}$ in $(\mathbb{R}, \mathcal{T}_{\text{std}})$ is:

- | | |
|----------------|-----------------|
| (A) A itself | (C) $\emptyset$ |
| (B) $\{0\}$ | (D) $[0, 1]$ |

✓ **Answer: (C)** No open interval (a, b) is contained in the countable set A , so $\overset{\circ}{A} = \emptyset$. The set has no interior points in $\mathbb{R}$.

Q5. If $\mathcal{T}_1$ and $\mathcal{T}_2$ are topologies on X , then $\mathcal{T}_1 \cap \mathcal{T}_2$:

- | | |
|---------------------------|---|
| (A) May not be a topology | (C) Is the discrete topology |
| (B) Is always a topology | (D) Equals $\mathcal{T}_1 \cup \mathcal{T}_2$ |

✓ **Answer: (B)** The intersection of any family of topologies on X is again a topology. However, the *union* of two topologies need not be a topology.

Q6. The closure of $(0, 1)$ in $(\mathbb{R}, \mathcal{T}_{\text{std}})$ is:

- | | |
|--------------|--------------|
| (A) $(0, 1)$ | (C) $(0, 1]$ |
| (B) $[0, 1)$ | (D) $[0, 1]$ |

✓ **Answer: (D)** The smallest closed set containing $(0, 1)$ is $[0, 1]$. The limit points 0 and 1 are added to get $\overline{(0, 1)} = [0, 1]$.

2. UNIT 2: SUBSPACE, PRODUCT & QUOTIENT TOPOLOGIES

2.1 Subspace Topology

Subspace (Induced / Relative) Topology

Let $(X, \mathcal{T})$ be a topological space and $Y \subseteq X$ a non-empty subset. The **subspace topology** on Y is:

$$\mathcal{T}_Y = \{U \cap Y : U \in \mathcal{T}\}$$

Open sets in Y are intersections of open sets in X with Y .

$(Y, \mathcal{T}_Y)$ is called a **subspace** (or **topological subspace**) of $(X, \mathcal{T})$.

Key Properties of Subspaces

- F is closed in Y iff $F = C \cap Y$ for some closed set C in X
- If $Z \subseteq Y \subseteq X$, the subspace topology on Z from Y equals the subspace topology on Z from X
- If Y is open in X : V open in $Y \Rightarrow V$ open in X
- If Y is closed in X : F closed in $Y \Rightarrow F$ closed in X
- **Hereditary properties:** T_1, T_2 , metrizability are inherited by subspaces

LaTeX Code

```
\mathcal{T}_Y = \{ U \cap Y : U \in \mathcal{T} \}
```

Example: Subspace Topology

$X = \mathbb{R}$ with standard topology, $Y = [0, 2]$.

Open set $U = (0.5, 1.5) \in \mathcal{T}_{\mathbb{R}} \Rightarrow U \cap Y = (0.5, 1.5) \in \mathcal{T}_Y$.

Open set $U = (-1, 1) \in \mathcal{T}_{\mathbb{R}} \Rightarrow U \cap Y = [0, 1) \in \mathcal{T}_Y$.

So $[0, 1)$ is **open in Y** even though it is not open in $\mathbb{R}$.

2.2 Product Topology

Product Topology (Finite and Infinite)

Finite product: For $(X_1, \mathcal{T}_1)$ and $(X_2, \mathcal{T}_2)$, the **product topology** on $X_1 \times X_2$ has basis:

$$\mathcal{B} = \{U_1 \times U_2 : U_1 \in \mathcal{T}_1, U_2 \in \mathcal{T}_2\}$$

Arbitrary product (Tychonoff topology): For $\{(X_\alpha, \mathcal{T}_\alpha)\}_{\alpha \in A}$, the product topology on $\prod_\alpha X_\alpha$ is generated by the sub-basis:

$$\mathcal{S} = \{\pi_\alpha^{-1}(U_\alpha) : \alpha \in A, U_\alpha \in \mathcal{T}_\alpha\}$$

where $\pi_\alpha : \prod X_\alpha \rightarrow X_\alpha$ are the **projection maps**.

Properties of Product Topology

- Projection maps π_α are continuous and open
- A map $f : Y \rightarrow \prod X_\alpha$ is continuous iff each $\pi_\alpha \circ f : Y \rightarrow X_\alpha$ is continuous (**Universal Property**)
- Product of Hausdorff spaces is Hausdorff
- Product of connected spaces is connected
- **Tychonoff Theorem:** Product of compact spaces is compact (see Unit 5)

LaTeX Code

```
X_1 \times X_2, \quad \mathcal{B} = \{U_1 \times U_2 : U_1 \in \mathcal{T}_1, \\ U_2 \in \mathcal{T}_2\} \\ \pi_{\alpha}^{-1}(U_{\alpha}), \quad \prod_{\alpha} X_{\alpha}
```

2.3 Quotient Topology**Quotient Topology**

Let $(X, \mathcal{T})$ be a space, $\sim$ an equivalence relation on X , and $q : X \rightarrow X/\sim$ the quotient map. The **quotient topology** on $X/\sim$ is:

$$\mathcal{T}_{X/\sim} = \{V \subseteq X/\sim : q^{-1}(V) \in \mathcal{T}\}$$

q is continuous, and it is the **finest** topology on $X/\sim$ making q continuous.

2.4 MCQs: Subspace, Product, Quotient Topologies

Q1. Let $Y = [0, 1] \subset \mathbb{R}$. Which set is open in Y but NOT open in $\mathbb{R}$?

(A) $(0, 1)$

(C) $(0.3, 0.7)$

(B) $[0, 0.5]$

(D) $\mathbb{R}$

✓ **Answer: (B)** $[0, 0.5) = (-1, 0.5) \cap [0, 1] \in \mathcal{T}_Y$. But $[0, 0.5)$ is not open in $\mathbb{R}$ since no open interval around 0 is contained in $[0, 0.5)$.

Q2. The projection map $\pi_1 : X_1 \times X_2 \rightarrow X_1$ in the product topology is:

- | | |
|-------------------------------|---------------------|
| (A) A closed map | (C) Discontinuous |
| (B) A continuous and open map | (D) A homeomorphism |

✓ **Answer: (B)** Projection maps in the product topology are always **continuous** (by definition of product topology) and **open** (preimage of open sets are open cylinders), but not closed in general.

Q3. A map $f : Y \rightarrow X_1 \times X_2$ is continuous iff:

- | | |
|---|-----------------------|
| (A) f is bijective | (C) $f(Y)$ is compact |
| (B) $\pi_1 \circ f$ and $\pi_2 \circ f$ are both continuous | (D) Y is Hausdorff |

✓ **Answer: (B)** By the **Universal Property** of product topology: f is continuous iff each coordinate function $\pi_i \circ f$ is continuous. This is the characterising property of the product topology.

Q4. The product topology on $\mathbb{R}^n = \mathbb{R} \times \mathbb{R} \times \cdots \times \mathbb{R}$ (n times) coincides with:

- | | |
|-------------------------|---------------------------------|
| (A) Discrete topology | (C) Standard Euclidean topology |
| (B) Indiscrete topology | (D) Cofinite topology |

✓ **Answer: (C)** The product topology on $\mathbb{R}^n$ (finite product of $\mathbb{R}$ with standard topology) equals the **standard Euclidean topology** induced by the Euclidean metric.

Q5. In the quotient topology, the quotient map $q : X \rightarrow X/\sim$ is:

- | | |
|-------------------|----------------------------|
| (A) Always open | (C) Always continuous |
| (B) Always closed | (D) Always a homeomorphism |

✓ **Answer: (C)** The quotient map is *always* continuous by definition of the quotient topology. It need not be open or closed.

3. UNIT 3: CONTINUITY AND HOMEOMORPHISM

3.1 Continuous Maps

Continuity in Topological Spaces

A function $f : (X, \mathcal{T}_X) \rightarrow (Y, \mathcal{T}_Y)$ is **continuous** if:

$$\forall V \in \mathcal{T}_Y, \quad f^{-1}(V) \in \mathcal{T}_X$$

That is, the preimage of every open set in Y is open in X .

Equivalent formulations:

1. Preimage of every closed set is closed
2. For each $x \in X$ and neighbourhood V of $f(x)$, $f^{-1}(V)$ is a neighbourhood of x
3. $f(\overline{A}) \subseteq \overline{f(A)}$ for all $A \subseteq X$
4. $f^{-1}(\overset{\circ}{B}) \subseteq f^{-1}(\overset{\circ}{f(B)})$ for all $B \subseteq Y$

LaTeX Code

```
f: (X, \mathcal{T}_X) \to (Y, \mathcal{T}_Y)
\forall V \in \mathcal{T}_Y, \quad f^{-1}(V) \in \mathcal{T}_X
f(\overline{A}) \subseteq \overline{f(A)}
```

Theorems on Continuous Maps

1. **Composition:** If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are continuous, then $g \circ f : X \rightarrow Z$ is continuous
2. **Restriction:** If $f : X \rightarrow Y$ is continuous and $A \subseteq X$, then $f|_A : A \rightarrow Y$ is continuous
3. **Pasting Lemma:** If $X = A \cup B$ where A, B are both closed (or both open), and $f|_A, f|_B$ are continuous and agree on $A \cap B$, then $f : X \rightarrow Y$ is continuous
4. Constant maps and identity maps are always continuous

3.2 Homeomorphism

Homeomorphism — Topological Equivalence

A bijection $f : X \rightarrow Y$ is a **homeomorphism** if both f and f^{-1} are continuous. Equivalently, f is a **bicontinuous bijection**.

$(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ are **homeomorphic** (written $X \cong Y$) if a homeomorphism exists.

Topological invariants (preserved by homeomorphism):

T_0, T_1, T_2 (Hausdorff), compactness, connectedness, path-connectedness

Homeomorphism vs Continuous Bijection

f bijective + f continuous $\not\Rightarrow f^{-1}$ continuous

Counter-example: $f : [0, 1) \rightarrow S^1$ defined by $f(t) = e^{2\pi it}$ is a continuous bijection but NOT a homeomorphism (since f^{-1} is not continuous at $f(0) = 1$).

However: **A continuous bijection from a compact space to a Hausdorff space IS a homeomorphism.**

LaTeX Code

```
f: X \to Y \text{\text{ homeomorphism}}, \quad X \cong Y
f \text{\text{ bicontinuous bijection}}, \quad f, f^{-1} \text{\text{ continuous}}
```

Classic Homeomorphism Examples

- $\mathbb{R} \cong (0, 1)$ via $f(x) = \frac{x}{1+|x|}$ or $f(x) = \frac{2}{\pi} \arctan(x)$
- Any open interval $(a, b) \cong \mathbb{R}$
- $S^1 \setminus \{p\} \cong \mathbb{R}$ (circle minus a point is homeomorphic to $\mathbb{R}$)
- $\mathbb{R}^n \cong$ open n -ball (open unit ball)
- A square $\cong$ a circle (both are compact 1-manifolds without boundary)
- **NOT homeomorphic:** $[0, 1]$ and $(0, 1)$ (compactness differs); $\mathbb{R}$ and $\mathbb{R}^2$ (connectedness after removing a point)

3.3 MCQs: Continuity and Homeomorphism

Q1. $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous (topologically) iff:

- (A) f is differentiable (C) f is increasing
 (B) Preimage of every open set is open (D) f is bounded

✓ **Answer: (B)** The topological definition: f is continuous iff $f^{-1}(V)$ is open in $\mathbb{R}$ for every open set $V \subseteq \mathbb{R}$. This is equivalent to the ε - δ definition.

Q2. The identity map $\text{id} : (X, \mathcal{T}_{\text{disc}}) \rightarrow (X, \mathcal{T}_{\text{indisc}})$ is:

- (A) Continuous and a homeomorphism (C) Not continuous
 (B) Continuous but not a homeomorphism (D) A closed map

✓ **Answer: (B)** id maps open sets (everything) in the discrete topology to the indiscrete topology where only $\emptyset, X$ need to have open preimages these do. But $\text{id}^{-1} : (X, \mathcal{T}_{\text{indisc}}) \rightarrow (X, \mathcal{T}_{\text{disc}})$ is not continuous (e.g., singletons are open in discrete but their preimage $\{x\}$ is not open in indiscrete). Not a homeomorphism.

Q3. $[0, 1]$ and $(0, 1)$ are NOT homeomorphic because:

- (A) One is bounded, the other is not (C) $(0, 1)$ has more elements
 (B) $[0, 1]$ is compact, $(0, 1)$ is not (D) They have different cardinalities

✓ **Answer: (B)** **Compactness** is a topological invariant. $[0, 1]$ is compact (closed and bounded in $\mathbb{R}$) while $(0, 1)$ is not compact. Since homeomorphisms preserve compactness, they cannot be homeomorphic.

Q4. A homeomorphism is:

- (A) A continuous bijection (C) A bicontinuous bijection
 (B) An isometry (D) A linear isomorphism

✓ **Answer: (C)** A homeomorphism requires: (1) bijection, (2) f continuous, (3) f^{-1} continuous. Option (A) is insufficient—a continuous bijection need not be a homeomorphism.

Q5. Which pair of spaces ARE homeomorphic?

- (A) $[0, 1]$ and $\mathbb{R}$ (C) $(0, 1)$ and $\mathbb{R}$
 (B) S^1 and $\mathbb{R}$ (D) $[0, 1]$ and $[0, 1]^2$

✓ **Answer: (C)** $(0, 1) \cong \mathbb{R}$ via $f(x) = \tan(\pi(x - \frac{1}{2}))$. The other pairs differ in compactness or connectedness properties.

4. UNIT 4: SEPARATION AXIOMS

Separation Axioms Overview

Separation axioms describe how well distinct points and sets can be “separated” by open sets.

Hierarchy:

$$T_0 \subset T_1 \subset T_2 \text{ (Hausdorff)} \subset T_{2\frac{1}{2}} \subset T_3 \text{ (Regular)} \subset T_{3\frac{1}{2}} \text{ (Tychonoff)} \subset T_4 \text{ (Normal)}$$

4.1 T_0 through T_4 Axioms

Separation Axioms Definitions

- T_0 (**Kolmogorov**): For any two distinct points $x \neq y$, there exists an open set containing one but not the other.
- T_1 (**Fréchet**): For any two distinct points $x \neq y$, there exist open sets $U \ni x$, $V \ni y$ with $y \notin U$ and $x \notin V$.

$$\Leftrightarrow \text{Every singleton } \{x\} \text{ is a closed set}$$

- T_2 (**Hausdorff**): For any $x \neq y$, there exist **disjoint** open sets $U \ni x$, $V \ni y$:

$$U \cap V = \emptyset$$

- T_3 (**Regular**): T_1 + for any closed set F and point $x \notin F$, there exist disjoint open sets separating x and F .
- T_4 (**Normal**): T_1 + for any two disjoint closed sets F_1, F_2 , there exist disjoint open sets $U_1 \supseteq F_1, U_2 \supseteq F_2$:

$$F_1 \cap F_2 = \emptyset \implies \exists U_1, U_2 \text{ open: } F_1 \subseteq U_1, F_2 \subseteq U_2, U_1 \cap U_2 = \emptyset$$

LaTeX Code

```
T_0:  x \neq y \Rightarrow \exists U \in \mathcal{T}: x \in U, y \notin U
T_2:  U \cap V = \emptyset, \quad x \in U, y \in V
T_4:  F_1 \cap F_2 = \emptyset \Rightarrow F_1 \subseteq U_1, F_2 \subseteq U_2, U_1 \cap U_2 = \emptyset
```

Key Theorems on Separation

1. **Urysohn's Lemma** (T_4): X is normal iff for any two disjoint closed sets A, B , there exists a continuous $f : X \rightarrow [0, 1]$ with $f(A) = 0$ and $f(B) = 1$.

$$A \cap B = \emptyset, A, B \text{ closed} \implies \exists f : X \rightarrow [0, 1] \text{ continuous}$$

2. **Tietze Extension Theorem**: X is normal iff every continuous $f : A \rightarrow [0, 1]$ on a closed subset A extends to a continuous $\tilde{f} : X \rightarrow [0, 1]$.

3. Every metric space is T_4 (normal).
4. Every compact Hausdorff space is T_4 (normal).
5. Finite T_1 spaces have the discrete topology.

LaTeX Code — Urysohn's Lemma

```
\exists f: X \to [0,1] \text{ continuous, } f(A)=\{0\}, f(B)=\{1\} \\ A \cap B = \emptyset, \text{quad } A, B \text{ closed in normal } X
```

4.2 MCQs: Separation Axioms

Q1. In a T_1 space, every singleton set is:

- | | |
|------------|-----------------------------|
| (A) Open | (C) Clopen |
| (B) Closed | (D) Neither open nor closed |

✓ **Answer: (B)** A fundamental characterisation: X is T_1 if and only if every singleton $\{x\}$ is a **closed set**. Equivalently, finite sets are closed in T_1 spaces.

Q2. A topological space is Hausdorff (T_2) if:

- | | |
|---|--|
| (A) Every pair of points has a common neighbourhood | (C) Any two distinct points can be separated by disjoint open sets |
| (B) Every continuous function is bounded | (D) Every closed set is compact |

✓ **Answer: (C)** T_2 (Hausdorff) precisely means: $\forall x \neq y, \exists$ disjoint open sets $U \ni x, V \ni y$ with $U \cap V = \emptyset$. This is the most important separation axiom.

Q3. Which separation axiom is equivalent to: “disjoint closed sets can be separated by disjoint open sets” (with T_1)?

- | | |
|-----------|---------------------|
| (A) T_0 | (C) T_3 (Regular) |
| (B) T_1 | (D) T_4 (Normal) |

✓ **Answer: (D)** **Normal** (T_4) = T_1 + any two disjoint closed sets can be separated by disjoint open sets. T_3 separates a point from a closed set.

Q4. Urysohn's Lemma gives a continuous function $f : X \rightarrow [0, 1]$ separating disjoint closed sets in a _____ space.

- | | |
|-------------------------|----------------------|
| (A) Hausdorff (T_2) | (C) Normal (T_4) |
| (B) Regular (T_3) | (D) T_0 |

✓ **Answer: (C)** Urysohn's Lemma applies to **normal** (T_4) spaces. It states that disjoint closed sets in a normal space can be separated by a continuous real-valued function.

Q5. Every metric space satisfies which separation axiom?

- | | |
|----------------|--------------------|
| (A) T_0 only | (C) T_3 only |
| (B) T_2 only | (D) T_4 (Normal) |

✓ **Answer: (D)** Every metric space is T_4 (normal). In fact, metric spaces satisfy all separation axioms T_0 through T_4 . They are **perfectly normal**.

Q6. Which implication is CORRECT among separation axioms?

- | | |
|---|---------------------------|
| (A) $T_3 \Rightarrow T_4$ | (C) $T_0 \Rightarrow T_4$ |
| (B) $T_4 \Rightarrow T_3 \Rightarrow T_2 \Rightarrow T_1 \Rightarrow T_0$ | (D) $T_2 \Rightarrow T_4$ |

✓ **Answer: (B)** The correct hierarchy is $T_4 \Rightarrow T_3 \Rightarrow T_2 \Rightarrow T_1 \Rightarrow T_0$. Each higher axiom implies all lower ones.

5. UNIT 5: COMPACTNESS AND TYCHONOFF THEOREM

5.1 Compact Spaces

Compactness

A topological space X is **compact** if every open cover of X has a **finite subcover**.

An **open cover** of X is a collection $\{U_\alpha\}$ of open sets with $X = \bigcup_\alpha U_\alpha$.

A subset $K \subseteq X$ is compact if $(K, \mathcal{T}_K)$ (with subspace topology) is compact.

Compactness Definitions & Equivalences

For metric spaces (and subsets of $\mathbb{R}^n$), the following are equivalent:

1. K is compact (every open cover has a finite subcover)
2. K is sequentially compact (every sequence has a convergent subsequence)
3. **Heine-Borel Theorem** (for $\mathbb{R}^n$): K is compact $\iff K$ is **closed and bounded**

LaTeX Code

```
X = \bigcup_\alpha U_\alpha \implies \exists \text{ finite subcover } X =
U_{i_1} \cup \dots \cup U_{i_n}
K \text{ compact in } \mathbb{R}^n \iff K \text{ closed and bounded}
```

Key Theorems on Compactness

1. Closed subset of a compact space is compact
2. Compact subset of a Hausdorff space is closed
3. Continuous image of a compact space is compact
4. A continuous bijection from a compact space to a Hausdorff space is a homeomorphism
5. **Extreme Value Theorem:** Continuous $f : K \rightarrow \mathbb{R}$ on compact K attains its max and min
6. Compact Hausdorff $\implies$ Normal (T_4)
7. Finite union of compact sets is compact; arbitrary intersection of compact sets is compact

5.2 Tychonoff Theorem

Tychonoff Theorem (One of the Greatest Theorems in Topology)

An arbitrary product of compact spaces is compact (in the product topology).

$$\prod_{\alpha \in A} X_{\alpha} \text{ is compact } \iff \text{each } X_{\alpha} \text{ is compact}$$

Proof strategy: Uses **Alexander's Sub-basis Theorem** + Zorn's Lemma (or ultrafilter convergence).

Special cases:

- $[0, 1]^n$ is compact (finite product of $[0, 1]$)
- $[0, 1]^{\mathbb{N}}$ (Hilbert cube) is compact
- $[0, 1]^{\mathbb{R}}$ (uncountable product) is compact — consequence of Tychonoff

LaTeX Code — Tychonoff Theorem

```
\prod_{\alpha \in A} X_{\alpha} \text{ compact } \iff \text{each } X_{\alpha} \text{ compact}
\text{Equivalent to Axiom of Choice (AC)}
```

Finite Intersection Property (FIP)

A space X is compact iff it has the **Finite Intersection Property**: every collection of closed sets with non-empty finite intersections has non-empty total intersection:

$$X \text{ compact } \iff \forall \{F_{\alpha}\} \text{ closed: } \left(\bigcap_{i=1}^n F_{\alpha_i} \neq \emptyset \forall \text{ finite subcollection} \right) \Rightarrow \bigcap_{\alpha} F_{\alpha} \neq \emptyset$$

5.3 MCQs: Compactness & Tychonoff Theorem

Q1. Heine-Borel Theorem states that in $\mathbb{R}^n$, a set is compact iff it is:

- | | |
|--------------------------|--------------------------|
| (A) Open and bounded | (C) Closed and bounded |
| (B) Connected and closed | (D) Dense and metrizable |

✓ **Answer: (C)** The **Heine-Borel Theorem** (for $\mathbb{R}^n$): $K \subseteq \mathbb{R}^n$ is compact $\iff$ K is **closed and bounded**. This fails in infinite-dimensional spaces.

Q2. A continuous image of a compact space is:

- | | |
|------------|---------------|
| (A) Closed | (C) Compact |
| (B) Open | (D) Hausdorff |

✓ **Answer: (C)** If $f : X \rightarrow Y$ is continuous and X is compact, then $f(X)$ is **compact**. This is one of the most important properties of compactness.

Q3. Tychonoff Theorem states that the product of compact spaces is compact. This theorem is equivalent to:

- | | |
|-------------------------|------------------------------|
| (A) Zorn's Lemma | (C) The Continuum Hypothesis |
| (B) The Axiom of Choice | (D) Cauchy's Theorem |

✓ **Answer: (B)** Tychonoff Theorem (for arbitrary products) is equivalent to the **Axiom of Choice**. The finite product version does not require AC.

Q4. Is $[0, 1]^{\mathbb{N}}$ (countably infinite product) compact?

- | | |
|---|---|
| (A) No, infinite products are never compact | (C) Only if each factor is sequentially compact |
| (B) Yes, by Tychonoff Theorem since $[0, 1]$ is compact | (D) It depends on the metric |

✓ **Answer: (B)** By **Tychonoff Theorem**: since $[0, 1]$ is compact, *any* product $[0, 1]^A$ is compact for any index set A . This includes $[0, 1]^{\mathbb{N}}$ (the Hilbert cube).

Q5. A closed subset of a compact space is:

- | | |
|-----------------------------|-----------|
| (A) Not necessarily compact | (C) Open |
| (B) Compact | (D) Dense |

✓ **Answer: (B)** A **closed subset** of a compact space is always **compact**. (Proof: Given any open cover of F , add $X \setminus F$ to get a cover of X ; extract finite subcover; remove $X \setminus F$).

Q6. The unit circle S^1 is compact because:

- | | |
|--------------------------|---|
| (A) It is a metric space | (C) It is closed and bounded in $\mathbb{R}^2$, hence compact by Heine-Borel |
| (B) It is path-connected | (D) It is homeomorphic to $\mathbb{R}$ |

✓ **Answer: (C)** $S^1 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$ is closed (as a level set of continuous function) and bounded (contained in $[-1, 1]^2$). By **Heine-Borel**, it is compact.

6. UNIT 6: CONNECTEDNESS

6.1 Connected Spaces

Connectedness

A topological space X is **connected** if it cannot be written as a union of two non-empty disjoint open sets.

Equivalently, X is connected iff the only subsets that are both open and closed (clopen) are $\emptyset$ and X .

A subset $A \subseteq X$ is connected if $(A, \mathcal{T}_A)$ is connected.

Connected Space Conditions

X is **disconnected** iff $\exists$ non-empty disjoint open sets U, V with:

$$X = U \cup V, \quad U \cap V = \emptyset, \quad U \neq \emptyset, \quad V \neq \emptyset$$

Such a pair $\{U, V\}$ is called a **separation** of X .

Connected subsets of $\mathbb{R}$: exactly the intervals (bounded, unbounded, half-open, closed, open).

LaTeX Code

```
X = U \cup V, \quad U \cap V = \emptyset, \quad U \neq \emptyset, \quad V \neq \emptyset
```

Key Theorems on Connectedness

1. **Continuous image of a connected space is connected**
2. **Intermediate Value Theorem:** If $f : X \rightarrow \mathbb{R}$ is continuous, X connected, $f(a) = c$, $f(b) = d$, then f takes all values between c and d
3. If $\{A_\alpha\}$ are connected and $\bigcap_\alpha A_\alpha \neq \emptyset$, then $\bigcup_\alpha A_\alpha$ is connected
4. Closure of a connected set is connected: A connected $\Rightarrow \overline{A}$ connected
5. Product of connected spaces is connected
6. $\mathbb{R}^n$ is connected for all $n \geq 1$

6.2 Path-Connectedness

Path-Connectedness

X is **path-connected** if for any $x, y \in X$, there exists a continuous map (path) $\gamma : [0, 1] \rightarrow X$ with $\gamma(0) = x$ and $\gamma(1) = y$.

Relation: Path-connected $\Rightarrow$ Connected.

The converse is **FALSE**: the **topologist's sine curve**:

$$S = \left\{ \left(x, \sin \frac{1}{x} \right) : x > 0 \right\} \cup \{(0, 0)\}$$

is connected but **not** path-connected.

Components and Local Connectedness

Connected component: maximal connected subset of X .

Path component: maximal path-connected subset of X .

X is **locally connected** at x if every neighbourhood of x contains a connected open neighbourhood.

X is **locally path-connected** at x if every neighbourhood of x contains a path-connected open neighbourhood.

If X is locally path-connected: connected $\iff$ path-connected.

LaTeX Code

```
\gamma: [0,1] \to X, \quad \gamma(0)=x, \quad \gamma(1)=y
S=\left\{(x,\sin\tfrac{1}{x}):x>0\right\}\cup\{(0,0)\}
```

6.3 MCQs: Connectedness

Q1. Which subset of $\mathbb{R}$ is NOT connected?

- | | |
|------------------------------|--------------------|
| (A) $[0, 1]$ | (C) $(0, 5)$ |
| (B) $\mathbb{Q}$ (rationals) | (D) $(-\infty, 3]$ |

✓ **Answer: (B)** $\mathbb{Q}$ is **not connected** in $\mathbb{R}$. For example, $\mathbb{Q} = (\mathbb{Q} \cap (-\infty, \sqrt{2})) \cup (\mathbb{Q} \cap (\sqrt{2}, +\infty))$ gives a separation. Connected subsets of $\mathbb{R}$ are exactly the intervals.

Q2. If $f : X \rightarrow Y$ is continuous and X is connected, then:

- | | |
|-----------------------|----------------------------|
| (A) Y is connected | (C) $f(X)$ is connected |
| (B) $f(X)$ is compact | (D) f is a homeomorphism |

✓ **Answer: (C)** Continuous image of connected is connected: $f(X)$ (the image, not necessarily all of Y) is connected.

Q3. Path-connected implies connected. The converse is:

- | | |
|-----------------|--|
| (A) Always true | (C) True for locally path-connected spaces |
| (B) Never true | (D) False in general |

✓ **Answer: (D)** In general, connected $\not\Rightarrow$ path-connected. Counter-example: topologist's sine curve. However, for locally path-connected spaces, connected $\iff$ path-connected.

Q4. $\mathbb{R}^n \setminus \{0\}$ is connected for:

- | | |
|------------------|----------------|
| (A) $n = 1$ only | (C) $n \geq 2$ |
| (B) $n = 2$ only | (D) No n |

✓ **Answer: (C)** $\mathbb{R}^1 \setminus \{0\} = (-\infty, 0) \cup (0, \infty)$ is disconnected. For $n \geq 2$, $\mathbb{R}^n \setminus \{0\}$ is path-connected (hence connected) since you can always go around the origin.

Q5. The number of connected components of $\mathbb{Q}$ (rationals) in $\mathbb{R}$ is:

- | | |
|-------------------------|--------------------------|
| (A) 1 (it is connected) | (C) Countably infinite |
| (B) 2 | (D) Uncountably infinite |

✓ **Answer: (D)** Each singleton $\{q\}$ for $q \in \mathbb{Q}$ is itself a connected component (since $\mathbb{Q}$ is totally disconnected). There are **uncountably**... actually countably many rationals, so countably infinite components. Each point is its own component.

Q6. The Intermediate Value Theorem holds because $[a, b]$ is:

- | | |
|---------------------------|---------------------|
| (A) Compact and Hausdorff | (C) Connected |
| (B) Metrizable | (D) Locally compact |

✓ **Answer: (C)** IVT is a consequence of **connectedness**: if $f : [a, b] \rightarrow \mathbb{R}$ is continuous and $[a, b]$ is connected, then $f([a, b])$ is connected (an interval), forcing f to take all intermediate values.

7. UNIT 7: METRIC SPACES

7.1 Metric Space Fundamentals

Metric Space Definition

A **metric space** (X, d) consists of a set X and a function $d : X \times X \rightarrow \mathbb{R}$ satisfying:

1. $d(x, y) \geq 0$ and $d(x, y) = 0 \iff x = y$ (Positive definiteness)
2. $d(x, y) = d(y, x)$ (Symmetry)
3. $d(x, z) \leq d(x, y) + d(y, z)$ (Triangle inequality)

Standard Metrics on $\mathbb{R}^n$

For $x = (x_1, \dots, x_n), y = (y_1, \dots, y_n) \in \mathbb{R}^n$:

$$d_2(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2} \quad (\text{Euclidean / } \ell^2 \text{ metric})$$

$$d_1(x, y) = \sum_{i=1}^n |x_i - y_i| \quad (\text{Taxicab / } \ell^1 \text{ metric})$$

$$d_\infty(x, y) = \max_{1 \leq i \leq n} |x_i - y_i| \quad (\text{Chebyshev / } \ell^\infty \text{ metric})$$

$$d_{\text{disc}}(x, y) = \begin{cases} 0 & x = y \\ 1 & x \neq y \end{cases} \quad (\text{Discrete metric})$$

LaTeX Code

```
d(x,z)\le d(x,y)+d(y,z)\quad(\text{Triangle inequality})
d_2(x,y)=\sqrt{\sum_{i=1}^n(x_i-y_i)^2}
d_\infty(x,y)=\max_{1\le i\le n}|x_i-y_i|
```

7.2 Open Balls, Topology, Convergence

Metric Topology

Open ball: $B(x, r) = \{y \in X : d(x, y) < r\}$ for $r > 0$

Closed ball: $\overline{B}(x, r) = \{y \in X : d(x, y) \leq r\}$

The collection $\{B(x, r) : x \in X, r > 0\}$ forms a **basis** for the metric topology.

Convergence: $x_n \rightarrow x$ iff $d(x_n, x) \rightarrow 0$, i.e., $\forall \varepsilon > 0, \exists N : n \geq N \Rightarrow d(x_n, x) < \varepsilon$

Continuity (metric): $f : X \rightarrow Y$ continuous at x iff $\forall \varepsilon > 0, \exists \delta > 0 : d(x, y) < \delta \Rightarrow d(f(x), f(y)) < \varepsilon$

LaTeX Code

```
B(x,r)=\{y\in X: d(x,y)<r\}
\forall \varepsilon>0,\exists N: n\ge N\Rightarrow d(x_n,x)<\varepsilon
d(x,y)<\delta \Rightarrow d(f(x),f(y))<\varepsilon
```

7.3 Completeness and Baire Category Theorem

Cauchy Sequences and Completeness

A sequence $\{x_n\}$ in (X, d) is a **Cauchy sequence** if:

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} : m, n \geq N \Rightarrow d(x_m, x_n) < \varepsilon$$

(X, d) is **complete** if every Cauchy sequence converges in X .

Complete metric spaces: $\mathbb{R}^n$, $\mathbb{Z}$, $C[a, b]$ (with sup-norm), ℓ^p spaces, Hilbert spaces.

Incomplete: $\mathbb{Q}$ (e.g., $1, 1.4, 1.41, 1.414, \dots \rightarrow \sqrt{2} \notin \mathbb{Q}$), $(0, 1)$ (the sequence $1/n$).

Baire Category Theorem

Baire Category Theorem: If (X, d) is a **complete metric space** (or a locally compact Hausdorff space), then X is a **Baire space**:

$$X \neq \bigcup_{n=1}^{\infty} F_n \quad \text{where each } F_n \text{ is closed with empty interior}$$

Equivalently, the countable intersection of dense open sets is dense.

Applications:

- $\mathbb{R}$ is uncountable (cannot be countable union of singletons each with empty interior... since $\mathbb{R}$ is complete)
- Existence of nowhere differentiable continuous functions
- $\mathbb{Q}$ is NOT a complete metric space (it is a countable union of singletons = nowhere dense sets)

LaTeX Code — Baire Category Theorem

```
X \neq \bigcup_{n=1}^{\infty} F_n, \quad \text{where } \text{int}(F_n) = \emptyset \text{ for all } n
\text{Countable intersection of dense open sets is dense}
```

Compactness in Metric Spaces

In metric spaces, the following are all equivalent:

1. K is compact (every open cover has finite subcover)
2. K is sequentially compact (every sequence has convergent subsequence in K)
3. K is complete and totally bounded ($\forall \varepsilon > 0$, K covered by finitely many ε -balls)

7.4 MCQs: Metric Spaces

Q1. Which of the following is NOT a valid metric on $\mathbb{R}$?

- (A) $d(x, y) = |x - y|$ (C) $d(x, y) = \frac{|x-y|}{1+|x-y|}$
 (B) $d(x, y) = |x - y|^2$ (D) $d(x, y) = \sqrt{|x - y|}$

✓ **Answer: (B)** $d(x, y) = |x - y|^2$ fails the triangle inequality. For $x = 0, y = 1, z = 2$: $d(0, 2) = 4$ but $d(0, 1) + d(1, 2) = 1 + 1 = 2 < 4$. All other options are valid metrics.

Q2. A sequence in a metric space that is convergent is:

- (A) Always Cauchy but not vice versa (C) Cauchy only in complete spaces
 (B) Always Cauchy (D) Not necessarily Cauchy

✓ **Answer: (B)** Every convergent sequence is **always Cauchy** in any metric space. The converse (Cauchy $\Rightarrow$ convergent) holds only in **complete** metric spaces.

Q3. $\mathbb{Q}$ with the standard metric is NOT complete because:

- (A) It is countable (C) There exist Cauchy sequences in $\mathbb{Q}$ that converge to irrationals
 (B) $d(x, y) = |x - y|$ is not a metric on $\mathbb{Q}$ (D) $\mathbb{Q}$ is dense in $\mathbb{R}$

✓ **Answer: (C)** E.g., $x_n = \lfloor \sqrt{2} \cdot 10^n \rfloor / 10^n$ is Cauchy in $\mathbb{Q}$ but converges to $\sqrt{2} \notin \mathbb{Q}$. Completeness fails.

Q4. The open ball $B(0, 1)$ in $(\mathbb{R}^2, d_\infty)$ where $d_\infty(x, y) = \max(|x_1 - y_1|, |x_2 - y_2|)$ is:

- (A) A circle (C) An open square $(-1, 1)^2$
 (B) An open disk (D) A triangle

✓ **Answer: (C)** For d_∞ : $B(0, 1) = \{(x_1, x_2) : \max(|x_1|, |x_2|) < 1\} = (-1, 1) \times (-1, 1)$, which is an open **square**.

Q5. Baire Category Theorem applies to:

- (A) All topological spaces (C) Only finite spaces
 (B) Compact Hausdorff spaces and complete metric spaces (D) Locally connected spaces

✓ **Answer: (B)** The Baire Category Theorem applies to: (1) **Complete metric spaces** and (2) **Locally compact Hausdorff spaces**. These are called *Baire spaces*.

Q6. In a complete metric space, a subset K is compact iff it is:

- | | |
|------------------------|----------------------------------|
| (A) Open and bounded | (C) Complete and totally bounded |
| (B) Closed and bounded | (D) Dense and metrizable |

✓ **Answer: (C)** In a complete metric space, K is compact $\iff K$ is **complete and totally bounded**. (Heine-Borel's "closed and bounded" version holds specifically in $\mathbb{R}^n$, not in all metric spaces.)

8. COMPLETE LATEX FORMULA REFERENCE

Topology Core LaTeX Formulas

$$\begin{aligned}
 (X, \mathcal{T}): & \text{ Topological Space, } \quad \emptyset, X \in \mathcal{T} \\
 \mathcal{T}_Y = & \{U \cap Y : U \in \mathcal{T}\} \quad (\text{Subspace}) \\
 \mathcal{B} = & \{U_1 \times U_2 : U_1 \in \mathcal{T}_1, U_2 \in \mathcal{T}_2\} \quad (\text{Product basis}) \\
 \mathring{A} = & \bigcup \{U \in \mathcal{T} : U \subseteq A\}, \quad \overline{A} = \bigcap \{F : F \text{ closed}, A \subseteq F\} \\
 \partial A = & \overline{A} \setminus \mathring{A}
 \end{aligned}$$

All Distributions and Theorems Quick Reference

Concept	Statement / Formula
Homeomorphism	$f : X \rightarrow Y$ bicontinuous bijection; $X \cong Y$
Heine-Borel	$K \subseteq \mathbb{R}^n$ compact $\iff$ closed + bounded
Tychonoff	$\prod_{\alpha} X_{\alpha}$ compact $\iff$ each X_{α} compact
Urysohn Lemma	X normal, $A \cap B = \emptyset$ (closed) $\Rightarrow \exists f : X \rightarrow [0, 1]$ continuous
Baire Category	Complete metric space $\neq \bigcup F_n$ (each F_n nowhere dense)
IVT (Topology)	$f : X \rightarrow \mathbb{R}$ continuous, X connected $\Rightarrow f(X)$ is an interval
T_1	Every singleton closed
T_2	Distinct points have disjoint open neighbourhoods
T_4	Disjoint closed sets have disjoint open neighbourhoods

Master LaTeX Code Collection

```

% Topological space:
(X,\mathcal{T}),\quad\emptyset,X\in\mathcal{T}

% Basis condition:
x\in B_3\subseteq B_1\cap B_2

% Interior and closure:
\mathring{A},\quad\overline{A},\quad\partial A=\overline{A}\setminus\mathring{A}

```

```

% Homeomorphism:
 $f: X \cong Y \iff f, f^{-1} \text{ continuous bijections}$ 
% Separation axioms:
 $T_2: U \cap V = \emptyset, T_4: F_1 \cap F_2 = \emptyset$ 
% Compactness:
 $X = \bigcup_{\alpha} U_{\alpha} \Rightarrow \text{exists finite subcover}$ 
% Tychonoff:
 $\prod_{\alpha \in A} X_{\alpha} \text{ compact} \iff \text{each } X_{\alpha} \text{ compact}$ 
% Metric:
 $d(x, z) \leq d(x, y) + d(y, z)$ 
% Completeness:
 $\forall \epsilon > 0, \exists N: m, n \geq N \Rightarrow d(x_m, x_n) < \epsilon$ 
% Baire:
 $X \neq \bigcup_n F_n, \text{quad } \bigcap_n F_n = \emptyset$ 

```

9. MASTER QUICK REVISION TABLE

Topic	Key Result / Formula	PSC Exam Tip
Topology definition	$\emptyset, X \in \mathcal{T}$; closed under unions and finite intersections	Check ALL three axioms
Subspace topology	$\mathcal{T}_Y = \{U \cap Y : U \in \mathcal{T}\}$	Open in Y may not be open in X
Product topology	Sub-basis: $\pi_\alpha^{-1}(U_\alpha)$; basis: $U_1 \times U_2$	Universal property key for continuity
Homeomorphism	Bicontinuous bijection; $X \cong Y$	Topological invariants are preserved
Hausdorff (T_2)	Distinct points $\Rightarrow$ disjoint open nbhds	Most natural separation axiom
Normal (T_4)	Disjoint closed sets $\Rightarrow$ disjoint open sets	Urysohn's Lemma applies here
Compactness	Every open cover has finite sub-cover	Heine-Borel for $\mathbb{R}^n$ only
Tychonoff Thm	Product of compacts is compact (uses AC)	Equivalent to Axiom of Choice
Connectedness	No separation $X = U \cup V$, disjoint open	Continuous image of connected is connected
Path-connected	$\exists \gamma : [0, 1] \rightarrow X$ continuous	Implies connected; converse FALSE
Completeness	Cauchy $\Rightarrow$ convergent in X	$\mathbb{Q}$ incomplete; $\mathbb{R}, \mathbb{R}^n$ complete
Baire Category	Complete metric space is Baire	$\mathbb{Q}$ not Baire; $\mathbb{R}$ is Baire
Metric equiv.	In metric: compact $\iff$ seq. compact $\iff$ complete + totally bounded	Heine-Borel is specific to $\mathbb{R}^n$

Table 1: Master Quick Revision: Topology for AJKPSC BPS-17 Mathematics Lecturer

10. 6-WEEK STUDY PLAN

Week	Topics	Daily Goals	Hours/Day
1	Topological spaces, open/-closed sets, basis, interior, closure, boundary	Master definitions; prove set is a topology; compute $\overset{\circ}{A}$, $\overline{A}$, ∂A	3–4 hrs
2	Subspace topology, product topology, quotient topology	Examples: $[0, 1] \subset \mathbb{R}$; product $\mathbb{R}^2$; 10 problems each construction	4 hrs
3	Continuity, homeomorphism, topological invariants	Prove/disprove homeomorphisms; master Pasting Lemma; classic counter-examples	4 hrs
4	Separation axioms T_0 – T_4 , Urysohn's Lemma, Tietze Extension	Classify spaces; prove Hausdorff properties; understand Urysohn construction	4 hrs
5	Compactness, Heine-Borel, Tychonoff Theorem, connectedness	Prove sets compact/connected; Tychonoff applications; IVT proofs	4 hrs
6	Metric spaces, completeness, Baire Category, revision + mock MCQs	Solve past AJKPSC papers; all metric problems; complete two full mock tests	4–5 hrs

Table 2: 6-Week AJKPSC BPS-17 Topology Study Schedule — Zamir Hussain

Important PSC Exam Tips for Topology

1. **Always check ALL three topology axioms** — a common MCQ trap is giving a collection missing one axiom.
2. $T_2 \Rightarrow T_1 \Rightarrow T_0$ (not the other way): Know counter-examples for each.
3. **Compactness in $\mathbb{R}^n$** : Use Heine-Borel (closed + bounded). This does NOT hold in general metric spaces.
4. **Tychonoff Theorem** is equivalent to the **Axiom of Choice** — a frequent exam question.
5. **Connected $\nRightarrow$ path-connected**: Topologist's sine curve is the key counter-example.
6. **Metric spaces**: Every metric space is Hausdorff (T_2), regular (T_3), and normal

(T_4) .

7. **Homeomorphism** requires BOTH f and f^{-1} continuous — a bijective continuous map need not be a homeomorphism.
8. **Subspace topology**: Open in subspace $\nRightarrow$ open in ambient space (e.g., $[0, 1)$ open in $[0, 2]$).
9. **Baire Category Theorem**: Complete metric spaces and locally compact Hausdorff spaces. $\mathbb{Q}$ is NOT a Baire space.
10. **AJKPSC pattern**: Definitions of compactness, Tychonoff Theorem, Hausdorff spaces, and metric completeness appear in every paper.

Exam Practice Resource

For further exam practice, the following resource provides a comprehensive collection of solved MCQs:

[Access the material here](#)

Prepared by: Zamir Hussain

BPS-17 PSC Mathematics Lecturer Examination | Topology | 2026 Edition

“In mathematics the art of proposing a question must be held of higher value than solving it.”

— Georg Cantor